

MOBILE MONEY LOAN ELIGIBILITY AND LOAN AMOUNT PREDICTION USING SUPERVISED AND UNSUPERVISED MACHINE LEARNING

A Machine Learning Project Report

Submitted in partial fulfilment of the requirements for the course assignment

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Project Area: Digital Credit Scoring and Mobile Money Analytics

Tools Used: Python, Jupyter Notebook, pandas, NumPy, scikit-learn, matplotlib and seaborn

Date: May 2026

IMAGE INSTRUCTION FOR TITLE PAGE

Optional: Insert a simple cover image showing a mobile phone, mobile money wallet, loan approval icon, and data analytics/AI symbols. Keep it professional and not too large.

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CHAPTER ONE

1. Introduction

Mobile money has become an important part of financial inclusion in Uganda because it allows people to send money, save, pay bills, and access digital financial services through a mobile phone. Digital credit services such as mobile money loans depend heavily on customer transaction behavior because many customers may not have traditional bank statements, salary slips, or collateral. This creates a need for intelligent data-driven systems that can analyze transaction patterns and estimate whether a customer is eligible for a loan and the amount that can be safely offered.

This project develops a hybrid machine learning approach for mobile money loan decision support. The system uses unsupervised learning to segment customers into risk groups and supervised learning to predict two outcomes: loan eligibility and eligible loan amount. The loan eligibility task is treated as a classification problem, while loan amount prediction is treated as a regression problem. The inclusion of clustering satisfies the unsupervised learning requirement and strengthens the project because it adds customer risk segmentation before supervised prediction.

The project is inspired by mobile money lending services such as MoKash, where mobile phone users can access savings and loan services through mobile money platforms. In such systems, customer activity, balances, transaction amounts, and financial behavior can support automated credit decisions. However, real lending systems require careful risk control, fairness, privacy protection, and validation using real repayment data.

FIGURE 1 INSTRUCTION

Insert a workflow diagram showing: Dataset -> Cleaning -> Feature Engineering -> Clustering -> Risk Grouping -> Classification Model -> Regression Model -> Final Loan Decision.

1.1 Problem Statement

Mobile money loan providers need to decide which customers should receive loans and how much they should receive. Traditional lending decisions may depend on manual assessment, formal employment records, collateral, or bank history, yet many mobile money customers do not have these records. This can exclude customers who are financially active but not formally banked.

At the same time, approving customers without proper risk assessment can lead to loan default and financial loss. A system that only uses simple rules may fail to detect complex patterns in customer behavior. Therefore, there is a need for a machine learning approach that can learn from transaction behavior, group customers according to risk, predict loan eligibility, and estimate a reasonable loan amount.

1.2 Significance of the Study

The study is significant because it demonstrates how machine learning can be applied to digital credit scoring in a mobile money context. It can help financial service providers make faster, more consistent, and more data-driven loan decisions. It also shows how unsupervised learning can support credit risk segmentation by grouping customers into low-risk, medium-risk, and high-risk categories.

For academic purposes, the project demonstrates practical use of classification, regression, clustering, preprocessing, feature engineering, model evaluation, and model comparison. It also provides a clear comparison of six supervised learning methods, making it suitable for a higher-level machine learning assignment.

1.3 Objectives

The main objective of this project is to develop a hybrid machine learning model for predicting mobile money loan eligibility and eligible loan amount.

The specific objectives are:

10. To clean and preprocess mobile money transaction data for machine learning.
11. To create customer-level behavioral features from transaction records.
12. To apply unsupervised clustering to group customers into risk categories.
13. To develop six supervised classification models for predicting loan eligibility.
14. To develop six supervised regression models for predicting eligible loan amount.
15. To compare the performance of the models and select the best-performing approach.
16. To interpret the results in relation to responsible mobile money lending.

1.4 Rationale

The rationale for using a hybrid machine learning approach is that loan decision-making involves both segmentation and prediction. Clustering helps discover natural customer groups without using labels, while supervised learning uses known or simulated targets to predict eligibility and amount. Combining these two approaches provides a stronger solution than using supervised learning alone, because the cluster label becomes an additional feature that summarizes customer risk behavior.

1.5 Background of the Problem

Mobile money has created new opportunities for financial inclusion in Uganda and other African countries. Many users perform frequent transactions through mobile wallets, including deposits, withdrawals, transfers, payments, and savings. These records can provide useful signals about financial activity, income patterns, balance behavior, and repayment capacity. Digital lenders can use such signals to offer small loans quickly, but they must also manage the risk of default.

In this project, the available dataset is treated as mobile money transaction data. Since the dataset may not contain real loan approval or repayment labels, academic target variables are created using transparent financial rules. The project therefore focuses on demonstrating machine learning methodology rather than building a real production lending system.

CHAPTER TWO

2. Literature Review

2.1 Introduction

This chapter reviews concepts related to digital credit scoring, supervised machine learning, unsupervised clustering, and model evaluation. The review supports the choice of methods used in the project and explains why multiple models are compared.

2.2 Digital Credit and Mobile Money Lending

Digital credit scoring uses customer data to estimate creditworthiness. In mobile money environments, transaction history can be useful because it shows customer activity and financial behavior. Useful indicators may include transaction frequency, average transaction amount, total inflow or outflow, balance changes, and cash-flow pressure. A customer with regular transactions and stable balances may be considered less risky than a customer with very high cash-flow pressure and unstable balances.

Mobile money credit systems should not only approve or reject customers but should also estimate a safe loan amount. Giving a loan that is too high may increase default risk, while giving a very low amount may reduce customer value. Therefore, credit scoring can be treated as two machine learning problems: classification for eligibility and regression for loan amount prediction.

2.3 Supervised Machine Learning for Credit Scoring

Supervised learning uses labelled data to train a model. In this project, supervised classification is used to predict whether a customer is eligible for a loan, while supervised regression is used to predict the amount the customer may qualify for. The supervised classification models used are Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, and K-Nearest Neighbors.

Logistic Regression is useful as a baseline classification model because it is simple and interpretable. Decision Trees are easy to understand because they use rule-based splits. Random Forest improves Decision Trees by building many trees and combining their predictions. Gradient Boosting builds models sequentially and focuses on correcting previous errors. Support Vector Machines can separate classes using decision boundaries, while K-Nearest Neighbors classifies a new customer by comparing the customer with similar customers in the training data.

2.4 Unsupervised Learning and Customer Segmentation

Unsupervised learning is used when the data does not have labels. In this project, clustering is used to group customers into risk categories. The clustering methods used are K-Means, Gaussian Mixture Model, and Agglomerative Clustering. K-Means separates customers into groups by minimizing within-cluster distance. Gaussian Mixture Model assumes that data comes

from a mixture of probability distributions. Agglomerative Clustering builds clusters through a hierarchical merging process.

The best clustering method is selected using Silhouette Score, Calinski-Harabasz Score, and Davies-Bouldin Score. A higher Silhouette Score generally indicates better separation between clusters, while a lower Davies-Bouldin Score indicates better compactness and separation.

2.5 Model Evaluation Metrics

Classification models are evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and Confusion Matrix. Accuracy shows the overall percentage of correct predictions. Precision is important in lending because it measures how many approved customers are actually eligible. Recall measures how many eligible customers were correctly identified. The F1 Score balances precision and recall. ROC-AUC measures the ability of the model to distinguish between eligible and non-eligible customers.

Regression models are evaluated using Mean Absolute Error, Root Mean Squared Error, and R-squared. MAE shows the average size of prediction errors. RMSE penalizes larger errors more strongly. R-squared shows how much variation in loan amount is explained by the model.

2.6 Summary

The literature and conceptual review show that machine learning is suitable for mobile money loan decision support because it can combine many behavioral indicators and detect patterns that simple rules may miss. The use of clustering improves the project by adding customer risk segmentation, while supervised models provide final eligibility and loan amount predictions.

CHAPTER THREE

3. Methodology

3.1 Research Design and Workflow

The project followed a quantitative experimental machine learning design. The workflow started with dataset loading, followed by cleaning, exploratory data analysis, customer-level feature engineering, unsupervised clustering, target creation, model training, model evaluation, comparison, and interpretation. The same processed dataset was used across models to ensure fair comparison.

FIGURE 1 INSTRUCTION

Use the workflow diagram from the notebook or create one in PowerPoint/draw.io showing the full machine learning pipeline.

3.2 Experimental Setup

3.2.1 Hardware Configuration

Table 1: Hardware configuration

Component	Specification
Processor	Personal computer CPU
Memory	At least 4 GB RAM recommended; 8 GB or more preferred
Storage	Enough storage for dataset and notebook files
GPU	Not required because the models are traditional machine learning models

3.2.2 Software Environment

Table 2: Software environment

Software/Library	Purpose
Python	Main programming language
Jupyter Notebook	Interactive model development environment
pandas	Data loading, cleaning, and manipulation
NumPy	Numerical computation
matplotlib and seaborn	Data visualization
scikit-learn	Machine learning algorithms and evaluation metrics

3.3 Dataset Description

The dataset used in the project contains mobile money-style transaction records. Important fields include transaction amount, origin balance before transaction, origin balance after transaction, destination balance before transaction, destination balance after transaction, transaction type, origin customer identifier, and destination customer identifier. The dataset is converted from transaction-level data into customer-level data so that loan decisions are made per customer rather than per transaction.

FIGURE 2 INSTRUCTION

Insert a screenshot of `df.info()` or the dataset columns and datatypes from the notebook. Caption it as: Figure 2: Dataset features and datatypes.

3.4 Data Exploration and Preprocessing

The preprocessing stage included checking dataset shape, viewing column names, identifying missing values, removing duplicates, converting numeric fields to correct formats, and filling missing values. Fraud-related columns, if available, were removed because the focus of the study is loan eligibility and loan amount prediction, not fraud detection. The dataset was also prepared for aggregation at customer level.

FIGURE 3 INSTRUCTION

Insert a screenshot or bar chart showing missing values before cleaning. Caption it as: Figure 3: Missing values

summary.

FIGURE 4 INSTRUCTION

Insert the histogram of log transaction amount generated in the notebook. Caption it as: Figure 4: Distribution of transaction amounts.

FIGURE 5 INSTRUCTION

Insert the correlation heatmap of numeric features generated in the notebook. Caption it as: Figure 5: Correlation heatmap of numeric features.

3.5 Feature Engineering

Feature engineering was performed to convert raw transactions into useful behavioral indicators. The project created customer-level features such as transaction count, average transaction amount, total transaction amount, maximum transaction amount, standard deviation of transaction amount, average origin balance, average new balance, balance change, cash-flow ratio, destination balance ratio, log transaction amount, and transaction type ratios.

Table 3: Main engineered features

Feature	Description
transaction_count	Number of transactions made by a customer
avg_transaction_amount	Average amount transacted by a customer
total_transaction_amount	Total value of customer transactions
avg_oldbalance_org	Average customer balance before transactions
avg_newbalance_orig	Average customer balance after transactions
avg_cash_flow_ratio	Ratio of transaction amount to available balance
cluster	Risk group label produced by unsupervised learning
risk_group_encoded	Numeric version of low, medium, or high risk group

3.6 Unsupervised Clustering

The unsupervised learning stage grouped customers into three clusters. The purpose was to discover customer segments without using loan labels. Three clustering methods were compared: K-Means, Gaussian Mixture Model, and Agglomerative Clustering. The selected number of clusters was three to represent low-risk, medium-risk, and high-risk customers.

Table 4: Unsupervised learning methods

Clustering Method	Reason for Use	Evaluation Metrics
K-Means	Simple and effective method for grouping customers by similar numeric behavior	Silhouette, Calinski-Harabasz, Davies-Bouldin
Gaussian Mixture Model	Allows probability-based soft clustering	Silhouette, Calinski-Harabasz, Davies-Bouldin
Agglomerative Clustering	Provides hierarchical grouping for comparison	Silhouette, Calinski-Harabasz, Davies-Bouldin

FIGURE 6 INSTRUCTION

Insert the PCA scatter plot of customer clusters from the notebook. Caption it as: Figure 6: Customer clusters visualized using PCA.

3.7 Creation of Target Variables

If the dataset does not contain real loan approval and repayment labels, the project creates academic target variables using transparent rules. A customer is marked as eligible when the customer has enough transaction activity, acceptable balance behavior, reasonable cash-flow pressure, and is not in the high-risk cluster. The eligible loan amount is then estimated from customer balance, transaction activity, and risk group, with a starting amount and a maximum cap.

Table 5: Target variables

Target Variable	Machine Learning Type	Description
is_eligible	Classification	Binary target showing whether the customer qualifies for a loan
eligible_loan_amount	Regression	Estimated loan amount the eligible customer can receive

3.8 Supervised Model Development

Two supervised learning tasks were developed. The first task was loan eligibility classification. The second task was loan amount regression. For classification, six supervised models were trained and compared. For regression, six supervised models were also trained and compared.

Table 6: Classification models

No.	Classification Model	Purpose
1	Logistic Regression	Baseline interpretable classification model
2	Decision Tree Classifier	Rule-based classification model
3	Random Forest Classifier	Ensemble tree-based classification model
4	Gradient Boosting Classifier	Boosted classification model
5	Support Vector Machine	Margin-based classification model
6	K-Nearest Neighbors Classifier	Similarity-based classification model

Table 7: Regression models

No.	Regression Model	Purpose
1	Linear Regression	Baseline numeric prediction model
2	Ridge Regression	Linear regression with regularization
3	Decision Tree Regressor	Rule-based loan amount prediction
4	Random Forest Regressor	Ensemble tree-based loan amount prediction
5	Gradient Boosting Regressor	Boosted loan amount prediction model

6	KNN Regressor	Similarity-based loan amount prediction
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3.9 Model Evaluation Strategy

The classification models were evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and Confusion Matrix. The regression models were evaluated using MAE, RMSE, and R-squared. The clustering methods were evaluated using Silhouette Score, Calinski-Harabasz Score, and Davies-Bouldin Score.

CHAPTER FOUR

4. Implementation and Testing

4.1 Introduction

This chapter explains how the project was implemented in Python using a Jupyter Notebook. The implementation followed the notebook pipeline from importing libraries to final prediction. The project was designed to be reproducible, meaning the notebook can be run again when the dataset is available.

4.2 Implementation Procedure

17. Import Python libraries and set random seed for reproducibility.
18. Load the mobile money transaction dataset from a CSV file.
19. Inspect columns, datatypes, missing values, and duplicate records.
20. Clean the dataset and convert numeric columns to correct formats.
21. Perform exploratory data analysis using graphs and summary statistics.
22. Create customer-level behavioral features.
23. Scale clustering features and apply three clustering algorithms.
24. Select the best clustering method and assign customer risk groups.
25. Create loan eligibility and loan amount target variables.
26. Train six classification models and compare their performance.
27. Train six regression models and compare their performance.
28. Select the best model and use it for final customer prediction.

4.3 Classification Model Training

The classification task predicted whether a customer was eligible for a loan. The input variables were customer-level features such as transaction count, transaction amount statistics, balance statistics, cash-flow ratio, cluster label, and risk group. The data was divided into training and testing sets. Models requiring scaling, such as Logistic Regression, SVM, and KNN, were trained using a pipeline with StandardScaler.

4.4 Regression Model Training

The regression task predicted eligible loan amount. Only customers classified as eligible were used for regression training because ineligible customers receive a loan amount of zero. The regression models were trained and compared using MAE, RMSE, and R-squared.

4.5 Hyperparameter Tuning

Basic hyperparameters were set to reduce overfitting and improve model reliability. For example, Random Forest used multiple estimators, Decision Trees used a maximum depth limit, and Logistic Regression used a higher maximum iteration value to support convergence. If more time and computing resources are available, GridSearchCV or RandomizedSearchCV can be used to tune parameters more deeply.

4.6 Final Prediction Function

The notebook includes a final prediction function that takes a customer record, predicts whether the customer is eligible, estimates the probability of eligibility, and returns the predicted loan amount. This provides a practical demonstration of how the trained models can be used for decision support.

CHAPTER FIVE

5. Results and Discussion

5.1 Introduction

This chapter presents the expected results format. Since results depend on running the notebook with the final dataset, the tables below should be completed using the outputs generated from the notebook. The discussion explains how to interpret the results after model execution.

5.2 Clustering Results

The clustering results compare K-Means, Gaussian Mixture Model, and Agglomerative Clustering. The best clustering method is selected mainly by the highest Silhouette Score and supported by the Calinski-Harabasz and Davies-Bouldin scores.

Table 8: Clustering model comparison

Clustering Method	Silhouette Score	Calinski-Harabasz Score	Davies-Bouldin Score	Rank
K-Means	Insert value from notebook	Insert value from notebook	Insert value from notebook	Insert rank
Gaussian Mixture Model	Insert value from notebook	Insert value from notebook	Insert value from notebook	Insert rank
Agglomerative Clustering	Insert value from notebook	Insert value from notebook	Insert value from notebook	Insert rank

After clustering, the clusters are profiled using mean values of transaction count, transaction amount, balance values, and cash-flow ratio. The cluster with the highest risk score is labelled High Risk, while the lowest is labelled Low Risk. The cluster label is then included as a feature in supervised prediction.

FIGURE 6 INSTRUCTION

Insert the PCA cluster plot showing low-risk, medium-risk and high-risk customer groups. Use the image generated by the notebook.

5.3 Loan Eligibility Classification Results

The classification models are compared using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. In a lending problem, the F1 Score and Precision are very important. Precision helps reduce the risk of approving customers who may not be suitable for loans.

Table 9: Classification model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	Insert value	Insert value	Insert value	Insert value	Insert value
Decision Tree	Insert value	Insert value	Insert value	Insert value	Insert value
Random Forest	Insert value	Insert value	Insert value	Insert value	Insert value
Gradient Boosting	Insert value	Insert value	Insert value	Insert value	Insert value
Support Vector Machine	Insert value	Insert value	Insert value	Insert value	Insert value
K-Nearest Neighbors	Insert value	Insert value	Insert value	Insert value	Insert value

FIGURE 7 INSTRUCTION

Insert the confusion matrix for the best classification model from the notebook. Caption it as: Figure 7: Confusion matrix for the best loan eligibility model.

5.4 Loan Amount Regression Results

The regression models are compared using MAE, RMSE, and R-squared. The best regression model is the one with the lowest RMSE and MAE and the highest R-squared value. RMSE is especially important because large loan amount errors can increase lending risk.

Table 10: Regression model comparison

Model	MAE	RMSE	R-squared
Linear Regression	Insert value	Insert value	Insert value
Ridge Regression	Insert value	Insert value	Insert value
Decision Tree Regressor	Insert value	Insert value	Insert value
Random Forest Regressor	Insert value	Insert value	Insert value
Gradient Boosting Regressor	Insert value	Insert value	Insert value
KNN Regressor	Insert value	Insert value	Insert value

FIGURE 8 INSTRUCTION

Insert the actual versus predicted loan amount scatter plot from the notebook. Caption it as: Figure 8: Actual versus predicted eligible loan amount.

5.5 Best Model Selection

The best classification model should be selected using F1 Score first, followed by Precision and ROC-AUC. The best regression model should be selected using RMSE first, followed by MAE and R-squared. If Random Forest or Gradient Boosting performs best, this is expected because tree-based ensemble models usually handle nonlinear relationships and feature interactions better than simple linear models.

Table 11: Summary of best models

Task	Best Model	Main Selection Metric	Reason
Clustering	Insert best clustering method	Silhouette Score	Best separation of customer groups
Loan Eligibility Classification	Insert best classifier	F1 Score and Precision	Best balance between correct approvals and risk control
Loan Amount Regression	Insert best regressor	RMSE and MAE	Lowest loan amount prediction error

FIGURE 9 INSTRUCTION

Insert the feature importance graph for the best tree-based model, if available. Caption it as: Figure 9: Top features influencing loan prediction.

5.6 Business Interpretation

The results should be interpreted from a financial risk perspective. A good loan model should not only produce high accuracy but should also avoid approving risky customers. The clustering output helps identify customer segments, while the classification output gives the approval decision. The regression output gives a loan amount that is linked to customer financial behavior. This makes the system useful as a decision-support tool for digital credit scoring.

CHAPTER SIX

6. Constraints and Ethical Considerations

6.1 Constraints

- The dataset may not contain real loan approval and repayment labels, so academic target variables may be simulated.
- The system is for academic demonstration and should not be deployed for real lending without real repayment data.

- The model may be affected by imbalance if one class has many more records than the other.
- The quality of predictions depends on the quality and completeness of transaction data.
- More customer information such as income, savings history, previous loans, and repayment behavior would improve the model.

6.2 Ethical Considerations

- Customer privacy must be protected by removing personal identifiers before analysis.
- The model should be tested for fairness to avoid excluding customers unfairly.
- Loan recommendations should support responsible lending and avoid pushing customers into over-indebtedness.
- Model decisions should be explainable, especially when a customer is rejected.
- The system should be used as decision support, not as the only basis for real financial decisions.

7. Conclusion

This project developed a hybrid machine learning framework for mobile money loan eligibility and loan amount prediction. The project combined unsupervised clustering with supervised classification and regression. Clustering was used to divide customers into low-risk, medium-risk, and high-risk groups, while classification predicted whether a customer was eligible for a loan and regression predicted the eligible loan amount.

The project compared three clustering algorithms, six classification models, and six regression models. The final model selection was based on objective evaluation metrics, including Silhouette Score for clustering, F1 Score and Precision for classification, and RMSE and MAE for regression. The project demonstrates that machine learning can support digital lending decisions by improving consistency, speed, and risk awareness.

8. Future Work

29. Use real loan repayment data to create more accurate loan eligibility labels.
30. Include previous loan history, default history, savings behavior, airtime usage, and customer demographics where legally and ethically allowed.
31. Apply advanced models such as XGBoost, LightGBM, CatBoost, or neural networks for comparison.
32. Use explainable AI methods such as SHAP or LIME to explain loan decisions.
33. Deploy the best model as a simple web or mobile application for demonstration.
34. Evaluate the model for fairness, bias, privacy, and responsible lending risks.

9. References

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Appendix A: Image and Screenshot Checklist

Table 12: Figures to add after running the notebook

Figure	What to Insert	Source in Notebook
Figure 1	Research workflow diagram	Create manually or use draw.io/PowerPoint
Figure 2	Dataset columns and datatypes	Output of <code>df.info()</code>
Figure 3	Missing values summary	Output of <code>df.isnull().sum()</code> or missing value bar chart
Figure 4	Transaction amount distribution	Histogram of log transaction amount
Figure 5	Correlation heatmap	Numeric correlation heatmap
Figure 6	PCA clustering scatter plot	PCA plot from clustering section
Figure 7	Confusion matrix	Best classification model confusion matrix
Figure 8	Actual vs predicted loan amount	Regression scatter plot
Figure 9	Feature importance	Best tree-based model feature importance plot